Probability from Observation:
Carathéodory and Stone Routes to the Observable Extension Theorem

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Abstract

We prove that a coherent family of observations determines a unique probability measure on the observable σ-algebra. The setting is a directed system of measurable outcome spaces with surjective refinement maps and a compatible family of finitely additive charges on the cylinder algebra. The necessary and sufficient condition for σ-additive extension is *collective exhaustion* (CE): no level can assign persistent mass to events the system as a whole sees as empty. CE is not derivable from any structural condition on the query system — a proved boundary, located precisely by the finite-cofinite counterexample and a model-theoretic argument via Łoś's theorem.

Two independent routes establish the result. The *Carathéodory route* assumes σ-additive marginals and constructs the global measure directly on the instantiation space, with no topological assumptions; CE is the condition the marginals satisfy. The *Stone route* assumes only finite additivity: compactness of the Stone space manufactures σ-additivity on the compact completion, and CE is then the support condition that descends the measure back to the instantiation space. Under shared hypotheses the two constructions coincide.

The Stone space $\mathrm{St}(\mathcal{C})$ is the canonical compact Hausdorff completion of the observable structure: the space of all consistent distinction patterns extending the cylinder algebra.

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1 Introduction

The programme of which this paper is a part does not begin with a probability space but with *distinctions*: the acts of observational separation by which an observer organises their experience of a system. State space, σ-algebra, and measure are not primitives — they are what coherent distinctions, pursued to their conclusion, are forced to produce. This paper proves the first step: a coherent family of observations determines a unique probability measure. The further steps — that this measure carries canonical dynamics, that the delay structure of observation (re)constructs the state space, and that finite data can certify (re)construction — are carried out in the companion papers.

What makes the first step possible is a single condition. *Collective exhaustion* (CE): A compatible family of finitely additive charges extends to a σ-additive measure if and only if, whenever a decreasing sequence of events vanishes globally, there exists a level at which the mass of the sequence converges to zero. CE rules out purely finitely additive charges, which assign persistent mass to sequences that vanish in the limit.

Two independent routes establish the result. The *Carathéodory route* (§4) assumes σ-additive marginals and constructs the global measure directly on the instantiation space. Sequential common refinements compress any countable cylinder family to a single level, reducing the σ-subadditivity check to a single level. By the CE theorem (§3), σ-additivity of the marginals is equivalent to CE.

The *Stone route* (§5) assumes only finite additivity and derives σ-additivity from compactness. Stone duality converts the directed Boolean system into an inverse system of compact Hausdorff spaces; a compatible family of charges assembles into a σ-additive measure on the inverse limit. Descending from the Stone space to Ω requires two conditions: Discriminability ensures the Stone embedding is injective, and CE appears as a support condition — the measure concentrates on the principal ultrafilters, i.e., on the image of Ω inside its Stone compactification.

CE is not derivable from any finitary or structural condition on the query system. The finite-cofinite content on $\mathbb{Q}$ satisfies every algebraic condition while failing CE; a model-theoretic argument via Łoś's theorem confirms this is a proved boundary, not a proof-search failure. CE is a non-derivable admissibility condition on charges: observational coherence alone does not suffice for probability.

The Stone space $\mathrm{St}(\mathcal{C})$, constructed as a technical device for measure extension, is the canonical compact Hausdorff completion of Ω for the observable topology (§6). CE's two faces — algebraic convergence and support on Ω — are the same condition seen from inside and outside the algebra. Its role as a completion of the state space is developed in the companion papers.

Organisation. Section 2 introduces query systems, cylinder sets, the observable σ-algebra, compatible charges, and the three hypotheses (Surjective Evaluation, Discriminability, CE). Section 3 establishes the CE theorem and the non-derivability of CE. Section 4 proves the Carathéodory route. Section 5 proves the Stone route. Section 6 draws the two routes together and discusses the Stone space. The main results have been formalized in Lean 4 using Mathlib; source files are available in the accompanying repository.

2 Setup: query systems and charges

2.1 Observational structure

Rather than beginning with a single σ-algebra of events, we work with a directed system of Boolean algebras representing observable distinctions at increasing levels of resolution; the global σ-algebra arises as their completion.

Definition 2.1 (Observational structure)**.** An *observational structure* consists of:

- a nonempty partially ordered set $(\iota, \leq)$, the *index set*;
- for each $i \in \iota$, a measurable space $(\mathsf{O}_i, \mathcal{O}_i)$, the *outcome space at level i*;
- for each $i \leq j$ in ι, a measurable surjection $\pi_{ij} : \mathsf{O}_j \to \mathsf{O}_i$, the *refinement map*,

satisfying $\pi_{ik} = \pi_{ij} \circ \pi_{jk}$ for all $i \leq j \leq k$.

Definition 2.2 (Instantiation; query system)**.** An *instantiation* of an observational structure $(\iota, \leq, \{(\mathsf{O}_i, \mathcal{O}_i)\}, \{\pi_{ij}\})$ is a set Ω together with maps $\mathrm{eval}_i : \Omega \to \mathsf{O}_i$ for each $i \in \iota$, the *evaluation maps*, satisfying the *coherence condition* $\pi_{ij} \circ \mathrm{eval}_j = \mathrm{eval}_i$ for all $i \leq j$. A *query system* is an observational structure together with an instantiation.

The index set, outcome spaces, and refinement maps encode the distinctions the observer can make and how finer levels coarsen to coarser ones. Choosing an instantiation Ω is an additional act: Ω is a set of states that realises the abstract structure concretely, and coherence says evaluation is compatible with coarsening.

2.2 Canonical instantiation and observables

Every observational structure admits a canonical instantiation, obtained by taking the space of all coherent assignments of outcomes across levels. This identifies Ω with the set of all globally consistent observation patterns; later, we show that a coherent family of charges determines a probability measure on this space. Explicitly, the canonical instantiation is the projective limit

$$\Omega = \varprojlim_{i\in\iota} \mathsf{O}_i = \Big\{\omega \in \prod_{i\in\iota} \mathsf{O}_i \;\Big|\; \pi_{ij}(\omega_j) = \omega_i \text{ whenever } i \le j\Big\},$$

with $\mathrm{eval}_i(\omega) = \omega_i$ and $\omega \in \Omega$ generic.

Example 2.3 (Bernoulli observations)**.** Let ι be the set of finite subsets of $\mathbb{N}$, ordered by inclusion. For $i \in \iota$, set $\mathsf{O}_i = \{0,1\}^{|i|}$ with its discrete σ-algebra $\mathcal{O}_i$, and for $i \subseteq j$ let $\pi_{ij} : \mathsf{O}_j \to \mathsf{O}_i$ be the coordinate projection. An instantiation assigns to each ω a consistent family of finite binary observations; the canonical instantiation identifies Ω with $\{0,1\}^{\mathbb{N}}$. Cylinder sets correspond to fixing finitely many coordinates, and a compatible family $\{\nu_i\}$ is precisely the system of finite-dimensional distributions of a Bernoulli process.

Definition 2.4 (Common coarsenings and common refinements)**.** The index set $(\iota, \le)$ admits *common coarsenings* if every pair $i, j \in \iota$ has a lower bound: there exists $k \le i$ and $k \le j$. It admits *common refinements* if every pair has an upper bound: there exists $k \ge i$ and $k \ge j$. It admits *sequential common refinements* if every countable family $\{i_n\}_{n\ge1} \subset \iota$ has an upper bound in ι.

Sequential common refinements imply common refinements. Common coarsenings ensure that cylinder sets form a π-system (Lemma 2.6 below); sequential common refinements supply the σ-subadditivity step in the Carathéodory route.

Definition 2.5 (Cylinder sets)**.** For $i \in \iota$ and $A \in \mathcal{O}_i$, the *level-i cylinder with base A* is

$$\mathrm{Cyl}(i, A) \;=\; \{\omega \in \Omega : \mathrm{eval}_i(\omega) \in A\}.$$

The *cylinder family* is $\mathcal{C} = \{\mathrm{Cyl}(i, A) : i \in \iota,\ A \in \mathcal{O}_i\}$, and the *observable σ-algebra* is $\sigma(\mathcal{C})$.

Lemma 2.6 (Cylinder π-system)**.** *Assume $(\iota, \le)$ admits common coarsenings. Then $\mathcal{C}$ is a π-system: it is closed under finite intersections.*

Proof. The coherence condition gives $\mathrm{Cyl}(i, A) = \mathrm{Cyl}(j, \pi_{ij}^{-1}(A))$ for $i \le j$: the same subset of Ω is a cylinder at any finer level. Given $\mathrm{Cyl}(i, A)$ and $\mathrm{Cyl}(j, B)$, let k be a common lower bound. Then

$$\mathrm{Cyl}(i, A) \cap \mathrm{Cyl}(j, B) = \mathrm{Cyl}(k,\ \pi_{ki}^{-1}(A) \cap \pi_{kj}^{-1}(B)),$$

which is again a cylinder. □

For each $i \in \iota$, let B_i denote the Boolean algebra of level-i cylinders $\{\mathrm{Cyl}(i, A) : A \in \mathcal{O}_i\}$; then $\mathcal{C} = \bigcup_{i\in\iota} B_i$. For $i \le j$, define the *connecting map*

$$\varphi_{ij} : B_i \to B_j, \qquad \varphi_{ij}(\mathrm{Cyl}(i, A)) = \mathrm{Cyl}(j, \pi_{ij}^{-1}(A)).$$

This is a well-defined Boolean algebra homomorphism. The family $\{B_i, \varphi_{ij}\}$ is a directed system of Boolean algebras.

2.3 Charges and hypotheses

Definition 2.7 (Compatible family)**.** A *charge* on a Boolean algebra B is a finitely additive function $\mu : B \to [0, \infty)$ with $\mu(\emptyset) = 0$. A family of charges $\{\mu_i\}_{i\in\iota}$ with μ_i on B_i is *compatible* if

$$\mu_j(\varphi_{ij}(E)) = \mu_i(E) \qquad \text{for all } i \le j \text{ and all } E \in B_i.$$

A compatible family also determines a charge on $\mathcal{C}$ by $\mu(\mathrm{Cyl}(i, A)) := \mu_i(\mathrm{Cyl}(i, A))$; compatibility ensures this is well-defined.

The main theorems require three conditions.

Definition 2.8 (Surjective Evaluation)**.** The query system satisfies *Surjective Evaluation* if $\mathrm{eval}_i : \Omega \to \mathsf{O}_i$ is surjective for every $i \in \iota$.

Surjective Evaluation is a structural condition ensuring that every outcome at every level is realized by some element of Ω; it implies that the refinement maps π_{ij} are surjective (by coherence of the evaluation maps) and that the connecting maps φ_{ij} are injective.

Definition 2.9 (Discriminability)**.** The query system *discriminates points* if for every $\omega \neq \omega'$ in Ω there exists $E \in \mathcal{C}$ with $\omega \in E$ and $\omega' \notin E$.

Discriminability ensures that the canonical embeddings remain injective.

Definition 2.10 (Collective Exhaustion)**.** A charge μ on $\mathcal{C}$ (the common extension of a compatible family $\{\mu_i\}$) satisfies *collective exhaustion* (CE) if: whenever $E_1 \supseteq E_2 \supseteq \cdots$ in $\mathcal{C}$ and $\bigcap_n E_n = \emptyset$, we have $\mu(E_n) \to 0$.

Exhaustion is the valuation-layer condition studied in §3: it rules out purely finitely additive charges and is the necessary and sufficient condition for σ-additive extensibility (see Example 2.11).

Example 2.11 (Finite–cofinite obstruction)**.** Let $\mathbb{N}$ carry the finite–cofinite algebra $\mathcal{E}$ where $A \in \mathcal{E}$ if A is finite or A^c is finite. Define a normalized charge by $\mu_0(A) = 0$ if A is finite and $\mu_0(A) = 1$ if A is cofinite.

Disjoint sets in $\mathcal{E}$ fall into exactly two cases: both finite, $\mu_0(A \cup B) = 0 + 0$, or one finite and one cofinite, $\mu_0(A \cup B) = 0 + 1$. Two disjoint cofinite sets cannot exist, thus finite additivity holds in any case.

Writing $\mathbb{N} = \bigcup_{n \geq 1}\{n\}$, each singleton has $\mu_0(\{n\}) = 0$ yet $\mu_0(\mathbb{N}) = 1$. Equivalently, the sequence $E_n = \{n, n+1, \ldots\}$ satisfies $E_n \downarrow \varnothing$, but each E_n is cofinite, so $\mu_0(E_n) = 1$ for all n: vanishing is not witnessed at any finite level, and mass is not exhausted. No σ-additive extension exists.

3 Collective Exhaustion

This section establishes collective exhaustion as the necessary and sufficient condition for σ-additive extensibility. We work first at the level of an abstract directed system of Boolean algebras, writing $\mathcal{E}_i$ for the algebra at level i and ℓ_i for its charge; the query-system interpretation returns in §§4–5.

3.1 Single-algebra extension

Definition 3.1 (Boolean charge space)**.** A *Boolean charge space* is a pair $(\mathcal{E}, \ell)$ where $\mathcal{E}$ is a Boolean algebra of subsets of a set O and $\ell : \mathcal{E} \to [0, 1]$ is a normalized finitely-additive valuation. No σ-algebra and no topology are assumed on O.

Under Stone's representation theorem [11], $\mathcal{E}$ is isomorphic to the clopen algebra of a compact totally-disconnected Hausdorff space $S(\mathcal{E})$, and $\sigma(\mathcal{E})$ is the Baire σ-algebra of $S(\mathcal{E})$. The sets in $\sigma(\mathcal{E}) \setminus \mathcal{E}$ are the Baire sets that are not clopen: events that countable operations can reach but that no single observable distinction can name.

Lemma 3.2 (Extension criterion)**.** *A normalized finitely-additive $\ell : \mathcal{E} \to [0, 1]$ extends to a σ-additive measure on $\sigma(\mathcal{E})$ if and only if ℓ is continuous at $\emptyset$: for every decreasing sequence $E_1 \supseteq E_2 \supseteq \cdots$ in $\mathcal{E}$ with $\bigcap_n E_n = \emptyset$ in $\sigma(\mathcal{E})$, we have $\ell(E_n) \to 0$. The extension is then unique.*

Proof. Standard; see Halmos [6] or Fremlin [5] Vol. 1. Continuity at $\emptyset$ is used in the Carathéodory extension argument to pass from finite additivity to σ-additivity on $\sigma(\mathcal{E})$. $\square$

The condition in Lemma 3.2 is not verifiable from within $\mathcal{E}$ alone: confirming $\bigcap_n E_n = \emptyset$ requires access to $\sigma(\mathcal{E})$. In a directed system, one might hope that a finer level $j \geq i$ could witness the emptiness that level i cannot see — but the same obstruction reappears: finite additivity at j does not force continuity at $\emptyset$ for sequences in $\sigma(\mathcal{E}_j)$. Something beyond any single level is required, and §3.2 shows the directed-system structure alone does not provide it.

3.2 Nontrivial refinement

Definition 3.3 (Nontrivial refinement)**.** Let $\pi : O_j \to O_i$ be a surjective refinement map, inducing the Boolean algebra homomorphism $\pi^{-1} : \mathcal{E}_i \to \mathcal{E}_j$. The refinement is *nontrivial* if there exists $A \in \mathcal{E}_j$ with $A \notin \pi^{-1}(\mathcal{E}_i)$.

Proposition 3.4 (Nontrivial refinement introduces new events)**.** *If $j \geq i$ is a nontrivial refinement, then $\pi^{-1}(\mathcal{E}_i) \subsetneq \mathcal{E}_j$. Equivalently, a level i at which $\mathcal{E}_j = \pi^{-1}(\mathcal{E}_i)$ for every $j \geq i$ is one above which the refinement order is algebraically trivial: no finer level adds new discriminative power.*

Proof. Nontriviality asserts the existence of $A \in \mathcal{E}_j$ with $A \notin \pi^{-1}(\mathcal{E}_i)$, directly giving the strict inclusion. The contrapositive is the stated equivalence. $\square$

Lemma 3.5 (Strict enlargement under σ-closure)**.** *Let $\mathcal{E}$ be a Boolean algebra of subsets of a countably infinite set O that is not a σ-algebra. Then $\sigma(\mathcal{E}) \setminus \mathcal{E} \neq \emptyset$.*

Proof. If $\mathcal{E}$ is not a σ-algebra, there exists a countable family $\{A_n\} \subset \mathcal{E}$ with $\bigcup_n A_n \notin \mathcal{E}$. By definition $\bigcup_n A_n \in \sigma(\mathcal{E})$, so it lies in $\sigma(\mathcal{E}) \setminus \mathcal{E}$. $\square$

3.3 CE characterises σ-additivity

We now state the directed-system version of CE and prove the main equivalence.

Definition 3.6 (Collectively exhaustive family)**.** A compatible family $\{\ell_i\}$ of normalized finitely-additive charges on a directed system is *collectively exhaustive* if: for every $i \in \mathcal{I}$ and every decreasing sequence $E_1 \supseteq E_2 \supseteq \cdots$ in $\mathcal{E}_i$ with $\bigcap_n E_n = \emptyset$, there exists $j \geq i$ such that $\ell_j(\pi_{ij}^{-1}(E_n)) \to 0$.

Here $\pi_{ij} : O_j \to O_i$ is the refinement map (j finer than i), so $\pi_{ij}^{-1}(E_n) \subseteq O_j$ is the preimage of $E_n \subseteq O_i$ at the finer level.
CE is a *valuation-layer* condition: it places a requirement on how the charges behave, not on the Boolean-algebraic index structure. It says that no level can assign persistent mass to events that the system as a whole sees as empty.

The following proposition is a separation result: it constructs the minimal directed system in which the index-layer hypotheses hold, while the finite-cofinite charge still fails σ-additive extension. The construction is a deliberate stress test, not a naturally arising observational system.

Proposition 3.7 (Index-layer conditions do not force σ-additivity)**.** *There exists a compatible family of normalized finitely-additive charges on a sequentially upper-directed directed system that fails σ-additive extension at every level.*

Proof. We take the simplest sequentially upper-directed index set: adjoin a single terminal element ω to $\mathbb{N}$, setting $n \leq \omega$ for all $n \in \mathbb{N}$. Take all outcome spaces to be $\mathbb{N}$, all event algebras to be the finite-cofinite algebra $\mathcal{E}$, all refinement maps to be the identity, and each ℓ_i to be the finite-cofinite charge ℓ of Example 2.11. Compatibility holds trivially. The system satisfies every structural condition. By Example 2.11, ℓ is not σ-additive and admits no σ-additive extension; hence every level fails σ-additive extension. $\square$

Theorem 3.8 (CE Characterisation Theorem)**.** *Let $\{\ell_i\}$ be a normalized compatible family of finitely-additive charges on a directed system indexed by $\mathcal{I}$. The following are equivalent:*

(i) The family is collectively exhaustive.

(ii) ℓ_i extends uniquely to a σ-additive measure on $\sigma(\mathcal{E}_i)$ for every $i \in \mathcal{I}$.

Proof. (i) $\Rightarrow$ (ii). Fix $i \in \mathcal{I}$ and a decreasing sequence $E_n \searrow \emptyset$ in $\mathcal{E}_i$. By CE, find $j \geq i$ with $\ell_j(\pi_{ij}^{-1}(E_n)) \to 0$. By compatibility, $\ell_i(E_n) = \ell_j(\pi_{ij}^{-1}(E_n)) \to 0$. So ℓ_i is continuous at $\emptyset$. Lemma 3.2 gives the unique σ-additive extension.

(ii) $\Rightarrow$ (i). Suppose ℓ_i extends to μ_i for every i. Let $E_n \searrow \emptyset$ in $\mathcal{E}_i$. Then $\ell_i(E_n) = \mu_i(E_n) \to \mu_i(\emptyset) = 0$ by σ-additivity of μ_i and normalization. Take $j = i$. $\square$

The question "do the structural conditions on a directed system force σ-additivity?" is ill-posed: it conflates conditions on the index structure with conditions on the charges. Proposition 3.7 separates them; Theorem 3.8 identifies the exact valuation-layer condition. Sequential upper-directedness enters only when assembling per-level extensions into a global measure (§4), not at the per-level step.

3.4 CE is non-derivable

The Yosida–Hewitt decomposition [13] identifies what kind of obstruction Proposition 3.7 has exposed. Every finitely additive probability on a Boolean algebra decomposes uniquely as $\ell = \ell_c + \ell_p$, where ℓ_c is σ-additive and ℓ_p is *purely finitely additive*. The finite-cofinite content ℓ of Example 2.11 is purely finitely additive, so $\ell_c = 0$ and $\ell_p = \ell$. By Theorem 3.8, CE holds precisely when ℓ extends to a σ-additive measure. In the language of Yosida–Hewitt, this is exactly the case in which no purely finitely additive remainder persists, i.e. $\ell_p = 0$. The obstruction to CE is therefore not a feature of the particular structural hypotheses chosen: it is the persistence of a purely finitely additive component that no first-order structural condition can eliminate. Proposition 3.9 below makes this precise.

Proposition 3.9 (CE non-derivability)**.** *Collective exhaustion is not implied by any structural condition on a query system.*

(i) (Particular case.) *Sequential upper-directedness, compatibility, and normalization do not imply CE.*

(ii) (Universal case.) *No condition Φ expressible in the first-order language of query systems implies CE.*

Proof. The particular case is Proposition 3.7: the finite-cofinite content satisfies all named structural hypotheses and fails CE.

For the universal case, we proceed in three steps.

Step 1. A first-order condition on query systems is preserved under ultraproducts by Łoś's theorem [9]: if Φ holds in every member of a family of query systems, it holds in their ultraproduct.

Step 2. Since the finite-cofinite construction works on any countably infinite set, we may realise it on $\mathbb{Q}$. Fix an enumeration $q : \mathbb{N} \to \mathbb{Q}$, let $\mathcal{U}$ be a nonprincipal ultrafilter on $\mathbb{N}$ extending the cofinite filter, and let δ_n be the point mass at $q(n) \in \mathbb{Q}$. Each δ_n is a normalized σ-additive content satisfying every structural condition including CE. The ultraproduct $\prod_{\mathcal{U}} \delta_n$ assigns to $E \subseteq \mathbb{Q}$ the value $\lim_{\mathcal{U}} \mathbf{1}_{q(n) \in E}$.

If E is finite then $q(n) \in E$ holds for only finitely many n, so the ultralimit is 0. If E is cofinite then $q(n) \in E$ holds for cofinitely many n, so the ultralimit is 1, since $\mathcal{U}$ extends the cofinite filter. The ultraproduct is therefore exactly the finite-cofinite content ℓ of Example 2.11 on $\mathbb{Q}$.

Step 3. By Step 1, any universally valid first-order condition Φ holds in the ultraproduct whenever it holds in each δ_n. But by Step 2 the ultraproduct is exactly ℓ, and ℓ fails CE. Therefore no first-order structural condition can imply CE. □

The ultrafilter-based content is the canonical witness: it passes every finite consistency test yet assigns unit mass to events whose infinite intersection is empty. CE is therefore the additional admissibility condition under which observational coherence furnishes probability, not a structural consequence of the query system.

4 Route 1: The Carathéodory extension

This section establishes the extension theorem by a direct Carathéodory argument. This is the third face of the argument: the measure constructed directly from within the algebra. The input is a query system satisfying Surjective Evaluation, sequential common refinements, and common coarsenings, together with a compatible family of σ-additive marginals. By Theorem 3.8, the σ-additivity of the marginals is equivalent to CE applied at each level.

4.1 Observational determination

Proposition 4.1 (Observational determination)**.** *Let the query system admit common coarsenings and let P, P' be probability measures on $(\Omega, \sigma(\mathcal{C}))$. If $(\mathrm{eval}_i)_\# P = (\mathrm{eval}_i)_\# P'$ for all $i \in \iota$, then $P = P'$.*

Proof. The equality of pushforward laws implies P and P' agree on every cylinder event. By Lemma 2.6 the cylinder family $\mathcal{C}$ is a π-system generating $\sigma(\mathcal{C})$. The π–λ theorem gives $P = P'$. □

4.2 Observable content

Lemma 4.2 (Finite observational content)**.** *Assume the query system admits common coarsenings and common refinements and satisfies Surjective Evaluation, and let $\{\nu_i\}_{i\in\iota}$ be an observationally compatible family with each $\nu_i \in \mathcal{P}(\mathsf{O}_i)$. Then the assignment*

$$\mu_0(\mathrm{Cyl}(i, A)) := \nu_i(A), \qquad A \in \mathcal{O}_i,$$

defines a well-defined finitely additive content on $\mathcal{C}$.

Proof. Well-definedness. Suppose $\mathrm{Cyl}(i, A) = \mathrm{Cyl}(j, B)$. Use common refinements to find k with $k \geq i$ and $k \geq j$. The constraint sets in O_k are $\pi_{ik}^{-1}(A)$ and $\pi_{jk}^{-1}(B)$. The cylinder equality and Surjective Evaluation (surjectivity of eval_k) imply these coincide in O_k. Compatibility then gives $\nu_i(A) = \nu_k(\pi_{ik}^{-1}(A)) = \nu_k(\pi_{jk}^{-1}(B)) = \nu_j(B)$.

Finite additivity. For disjoint cylinders at possibly different levels, compress to a common refinement k and use additivity of ν_k. □

4.3 Carathéodory extension theorem

Theorem 4.3 (Observational extension theorem)**.** *Suppose:*

(i) $(\iota, \leq)$ *admits common coarsenings and sequential common refinements;*

(ii) the query system satisfies Surjective Evaluation;

(iii) the family $\{\nu_i\}_{i\in\iota}$ is observationally compatible and each ν_i is a probability measure on $(\mathsf{O}_i, \mathcal{O}_i)$.

Then there exists a unique probability measure P on $(\Omega, \sigma(\mathcal{C}))$ such that $(\mathrm{eval}_i)_\# P = \nu_i$ for every $i \in \iota$.

Proof. Step 1: Finite content. Theorem 4.2 gives a well-defined finitely additive content μ_0 on $\mathcal{C}$.

Step 2: σ-subadditivity. Let $\mathrm{Cyl}(i, B) \subseteq \bigcup_{n\geq 1} \mathrm{Cyl}(i_n, A_n)$. Apply sequential common refinements to $\{i, i_1, i_2, \ldots\}$ to find a single index k with $k \geq i$ and $k \geq i_n$ for all n. Compress each cylinder to level k:

$$\mathrm{Cyl}(i, B) = \mathrm{Cyl}(k, E), \qquad \mathrm{Cyl}(i_n, A_n) = \mathrm{Cyl}(k, E_n),$$

where $E = \pi_{ik}^{-1}(B)$ and $E_n = \pi_{i_n k}^{-1}(A_n)$. Surjective Evaluation and the cylinder inclusion in Ω imply $E \subseteq \bigcup_n E_n$ in O_k. Therefore

$$\mu_0(\mathrm{Cyl}(i, B)) = \nu_k(E) \leq \sum_{n=1}^{\infty} \nu_k(E_n) = \sum_{n=1}^{\infty} \mu_0(\mathrm{Cyl}(i_n, A_n)),$$

using σ-subadditivity of the single measure ν_k.

Step 3: Carathéodory extension. The family $\mathcal{C}$ is a semiring generating $\sigma(\mathcal{C})$ and μ_0 is a σ-subadditive finitely additive content on it. Carathéodory's extension theorem [1, 2] applies and yields P on $(\Omega, \sigma(\mathcal{C}))$.

Step 4: Marginal recovery and normalization. For any $A \in \mathcal{O}_i$, $P(\mathrm{Cyl}(i, A)) = \mu_0(\mathrm{Cyl}(i, A)) = \nu_i(A)$, so $(\mathrm{eval}_i)_\# P = \nu_i$. Setting $A = \mathsf{O}_i$ gives $P(\Omega) = 1$.

Uniqueness. Two probability measures with the same evaluation marginals agree on $\mathcal{C}$, hence on $\sigma(\mathcal{C})$ by Theorem 4.1. □

Remark 4.4 (Relation to Kolmogorov's extension theorem)**.** Kolmogorov's theorem [8] constructs a probability measure on S^T from consistent finite-dimensional distributions. Theorem 4.3 plays the same role for query systems: the instantiation $\Omega = \varprojlim \mathsf{O}_i$ replaces the product S^T, compatible marginals replace consistent finite-dimensional distributions, and sequential common refinements replace the countable-cofinality condition. The key conceptual difference is that no topology is imposed on the outcome spaces: the measure P is built directly on the measurable structure of Ω by Carathéodory, without any appeal to tightness or compactness.

A third, topological route begins from finitely additive marginals and derives σ-additivity from compactness. Under *surjective projective uniform tightness* — for every $\varepsilon > 0$ there exist compact sets $K_i \subset \mathsf{O}_i$ with $\nu_i(\mathsf{O}_i \setminus K_i) < \varepsilon$ and $\pi_{ij}(K_j) = K_i$ for all $i \leq j$ — the Prokhorov theorem extends finitely additive marginals to a σ-additive global measure. This route assumes topological structure on the outcome spaces and is complementary to the Carathéodory and Stone routes, which require none.

5 Route 2: The Stone extension

This section gives the second, independent proof of the extension theorem. This is the fourth face: the same measure derived from outside, via compactness. The input is weaker: charges need only be finitely additive. σ-additivity is derived from the compactness of the Stone space.

The proof proceeds in four steps. Step A identifies the Stone space of the direct limit with the inverse limit of individual Stone spaces. The inverse limit measure step uses Choksi's theorem to produce a σ-additive measure on the inverse limit. Step B1 identifies the pullback of the Borel algebra of the Stone space with the observable σ-algebra. Step B2 shows that CE forces the measure to concentrate on the image of Ω.

5.1 Step A: Stone space of the direct limit

Lemma 5.1 (Injectivity of connecting maps)**.** *If the query system satisfies Surjective Evaluation, then $\varphi_{ij} : B_i \to B_j$ is an injective Boolean algebra homomorphism for every $i \leq j$.*

Proof. Surjective Evaluation and the coherence condition give $\pi_{ij}(\mathrm{eval}_j(\omega)) = \mathrm{eval}_i(\omega)$ for all ω, so $\mathrm{im}(\pi_{ij}) \supseteq \mathrm{im}(\mathrm{eval}_i) = \mathsf{O}_i$: the refinement maps are surjective. A surjective map π_{ij} has injective preimage map π_{ij}^{-1} on subsets, so $\varphi_{ij}(\mathrm{Cyl}(i, A)) = \mathrm{Cyl}(j, \pi_{ij}^{-1}(A))$ is injective on generators, hence on B_i. □

Proposition 5.2 (Step A)**.** *The Stone space of the direct limit satisfies* $\mathrm{St}\big(\bigcup_i B_i\big) \cong \varprojlim_i \mathrm{St}(B_i)$.

Proof. By Lemma 5.1, the connecting maps φ_{ij} are injective Boolean algebra homomorphisms. Under Stone duality, injective Boolean algebra homomorphisms correspond to surjective continuous maps between Stone spaces. The Stone space functor St converts colimits in the category of Boolean algebras to limits in the category of compact Hausdorff spaces [7, II.4]. Applied to $\bigcup_i B_i$ with its injective system, this gives $\mathrm{St}\big(\bigcup_i B_i\big) \cong \varprojlim_i \mathrm{St}(B_i)$. □

5.2 The inverse limit measure

Proposition 5.3 (Inverse limit measure)**.** *Let $\{\mu_i\}$ be a compatible family of normalised charges on $\{B_i\}$. Then there exists a unique probability measure $\hat{\mu}$ on $\mathcal{B}(\varprojlim_i \mathrm{St}(B_i))$ whose pushforward along each projection $\varprojlim_i \mathrm{St}(B_i) \to \mathrm{St}(B_j)$ agrees with the Borel measure on $\mathrm{St}(B_j)$ corresponding to μ_j.*

Proof. Step 1: Single-algebra correspondence. For each i, Stone duality identifies B_i with the clopen algebra $\mathrm{Clop}(\mathrm{St}(B_i))$. A charge μ_i on B_i thus defines a finitely additive set function on the clopens of the compact Hausdorff space $\mathrm{St}(B_i)$. Since $\mathrm{St}(B_i)$ is compact and totally disconnected, the clopens form a base for the topology and a finitely additive charge on them extends uniquely to a regular Borel measure; see [3], §6, or [6], §§53–54. Call this measure $\hat{\mu}_i$ on $\mathrm{St}(B_i)$.

Step 2: Compatibility. The compatibility condition $\mu_j(\varphi_{ij}(E)) = \mu_i(E)$ translates, under Stone correspondence, to $\hat{\mu}_i$ being the pushforward of $\hat{\mu}_j$ along the bonding map $\mathrm{St}(B_j) \to \mathrm{St}(B_i)$. The family $\{\hat{\mu}_i\}$ is compatible on the inverse system $\{\mathrm{St}(B_i)\}$.

Step 3: Extension to the inverse limit. Each $\mathrm{St}(B_i)$ is compact Hausdorff. The inverse limit $\varprojlim_i \mathrm{St}(B_i)$ is compact Hausdorff. By [4, Theorem 1], a compatible family of regular Borel probability measures on a cofiltered inverse system of compact Hausdorff spaces with surjective bonding maps extends uniquely to a regular Borel probability measure on the inverse limit. This gives $\hat{\mu}$ on $\mathcal{B}(\varprojlim_i \mathrm{St}(B_i))$. □

5.3 Step B1: the structural identification

Let $\mathrm{pure} : \Omega \to \mathrm{St}(\mathcal{C})$ denote the Stone embedding, which sends $\omega \in \Omega$ to its principal ultrafilter $\mathrm{pure}(\omega) = \{E \in \mathcal{C} : \omega \in E\}$. For $E \in \mathcal{C}$, write $[E] = \{u \in \mathrm{St}(\mathcal{C}) : E \in u\}$ for the corresponding basic clopen in $\mathrm{St}(\mathcal{C})$.

Proposition 5.4 (Step B1: structural identification)**.** *Suppose the query system discriminates points. Then*

$$\mathrm{pure}^{-1}\big(\mathcal{B}_{\mathrm{Ba}}(\mathrm{St}(\mathcal{C}))\big) \;=\; \sigma(\mathcal{C}),$$

where $\mathcal{B}_{\mathrm{Ba}}$ denotes the Baire σ-algebra.

Proof. **($\supseteq$):** For any $E = \mathrm{Cyl}(i, A) \in \mathcal{C}$, $\mathrm{pure}^{-1}([E]) = \{\omega : E \in \mathrm{pure}(\omega)\} = \{\omega : \omega \in E\} = E$. Since $[E]$ is clopen, hence Baire, every cylinder set is in $\mathrm{pure}^{-1}(\mathcal{B}_{\mathrm{Ba}})$, and therefore $\sigma(\mathcal{C}) \subseteq \mathrm{pure}^{-1}(\mathcal{B}_{\mathrm{Ba}})$.

($\subseteq$): The Stone space $\mathrm{St}(\mathcal{C})$ is compact and totally disconnected, so its clopens form a topological basis. The clopens are exactly the sets $[E]$ for $E \in \mathcal{C}$ [7, I.3.2], so the Baire σ-algebra of $\mathrm{St}(\mathcal{C})$ is $\sigma(\{[E] : E \in \mathcal{C}\})$. For any Baire set V, $\mathrm{pure}^{-1}(V) \in \sigma(\{\mathrm{pure}^{-1}([E]) : E \in \mathcal{C}\}) = \sigma(\mathcal{C})$.

Injectivity. Discriminability implies pure is injective: if $\omega \neq \omega'$, there exists $E \in \mathcal{C}$ with $\omega \in E$ and $\omega' \notin E$, so $E \in \mathrm{pure}(\omega)$ but $E \notin \mathrm{pure}(\omega')$. Injectivity ensures the pullback does not conflate distinct events. □

Remark 5.5. On a compact Hausdorff space the Baire σ-algebra is contained in the Borel σ-algebra, with equality when the space is second-countable. We work throughout with the Baire σ-algebra on $\mathrm{St}(\mathcal{C})$; this suffices because the measure $\hat{\mu}$ constructed above is a Baire measure (it is the unique extension of a charge on the clopen algebra), and Choksi's theorem applies to Baire measures on compact spaces.

5.4 Step B2: CE and the support condition

Proposition 5.6 (Step B2: support condition)**.** *Let μ be a charge on $\mathcal{C}$ satisfying CE. Then the corresponding Baire measure $\hat{\mu}$ on $\mathrm{St}(\mathcal{C})$ is supported on $\mathrm{pure}(\Omega)$, the set of principal ultrafilters.*

Proof. By the Yosida–Hewitt theorem [13, 10], every bounded charge on a Boolean algebra decomposes uniquely as $\mu = \mu_c + \mu_p$, where μ_c is countably additive and μ_p is purely finitely additive. Under Stone duality, μ_c corresponds to mass on the principal ultrafilters $\mathrm{pure}(\Omega) \subset \mathrm{St}(\mathcal{C})$, while μ_p corresponds to mass on the non-principal ultrafilters [10], Ch. 10.

CE is equivalent to $\mu_p = 0$: CE requires $\mu(E_n) \to 0$ whenever $E_n \searrow \emptyset$, which is exactly the condition that the purely finitely additive part vanishes (Theorem 3.8). Therefore $\hat{\mu}_p = 0$ and $\hat{\mu}$ is supported on $\mathrm{pure}(\Omega)$. □

5.5 The Stone extension theorem

Theorem 5.7 (Stone extension theorem)**.** *Let the query system satisfy Surjective Evaluation, Discriminability, and common coarsenings, and let $\{\mu_i\}$ be a compatible family of normalised charges on $\{B_i\}$ satisfying CE. Then there exists a unique probability measure P on $(\Omega, \sigma(\mathcal{C}))$ such that*

$$P(\mathrm{Cyl}(i, A)) \;=\; \mu_i(\mathrm{Cyl}(i, A)) \qquad \textit{for all } i \in \iota \textit{ and } A \in \mathcal{O}_i.$$

Proof. The proof assembles the preceding propositions:

$$\{\mu_i\} \xrightarrow{\S5.2,\,\text{Step 1}} \{\hat{\mu}_i\} \xrightarrow{\S5.2,\,\text{Step 3}} \hat{\mu} \text{ on } \varprojlim_i \mathrm{St}(B_i) \xrightarrow{\text{B1}} P \text{ on } \sigma(\mathcal{C}).$$

Existence. By Proposition 5.2, $\mathrm{St}(\mathcal{C}) \cong \varprojlim_i \mathrm{St}(B_i)$. By Proposition 5.3, the compatible family $\{\mu_i\}$ extends to a Baire probability measure $\hat{\mu}$ on $\varprojlim_i \mathrm{St}(B_i) \cong \mathrm{St}(\mathcal{C})$.

By Proposition 5.6, $\hat{\mu}$ is supported on $\mathrm{pure}(\Omega)$. Since pure is injective (Discriminability), the pushforward $P = (\mathrm{pure})_*\hat{\mu}$ — the image measure under pure^{-1} on the support — is a well-defined probability measure on Ω.

By Proposition 5.4, pure^{-1} maps Baire sets in $\mathrm{St}(\mathcal{C})$ to sets in $\sigma(\mathcal{C})$, so P is defined on $\sigma(\mathcal{C})$. For any cylinder set $E = \mathrm{Cyl}(i, A)$,

$$P(E) \;=\; \hat{\mu}([E]) \;=\; \mu_i(\mathrm{Cyl}(i, A)).$$

Uniqueness. Two probability measures on $(\Omega, \sigma(\mathcal{C}))$ agreeing on all cylinder sets are equal by Theorem 4.1, which requires common coarsenings. The Stone theorem therefore assumes common coarsenings in addition to Surjective Evaluation, Discriminability, and CE. □

It may appear that the compactness of the Stone space eliminates the need for CE. Compactness does ensure that $\hat{\mu}$ on $\mathrm{St}(\mathcal{C})$ is σ-additive. But $\hat{\mu}$ lives on the Stone space, not on Ω. The descent from $\mathrm{St}(\mathcal{C})$ to Ω requires that no mass escapes to the non-principal ultrafilters, and that is precisely CE (Proposition 5.6). In the Stone picture, CE is not an additional hypothesis but a *support condition*: the measure concentrates on the image of the sample space inside its Stone compactification.

6 The bridge

6.1 Two routes, one theorem

Under shared hypotheses — Surjective Evaluation, Discriminability, and σ-additive compatible marginals — both routes produce the same measure.

Proposition 6.1 (Coincidence of routes)**.** *Suppose the query system satisfies Surjective Evaluation, Discriminability, and sequential common refinements and common coarsenings, and let $\{\nu_i\}$ be a compatible family of probability measures. Let P^{C} be the measure produced by Theorem 4.3 and P^{S} the measure produced by Theorem 5.7. Then $P^{\mathrm{C}} = P^{\mathrm{S}}$.*

Proof. Both measures agree on every cylinder set: $P^{\mathrm{C}}(\mathrm{Cyl}(i, A)) = \nu_i(A) = P^{\mathrm{S}}(\mathrm{Cyl}(i, A))$. By observational determination (Theorem 4.1), they are equal on $\sigma(\mathcal{C})$. □

The two routes are genuinely independent proofs of the same theorem. The Carathéodory route works directly within measure theory; the Stone route takes a detour through topology, using compactness of the Stone space to derive σ-additivity that the Carathéodory route assumes. That they produce the same measure is not obvious from the constructions alone; it is a consequence of uniqueness (Theorem 4.1), which is doing real work here.

6.2 CE's two faces

CE appears in the two routes in structurally different forms, yet it is the same condition.

In the Carathéodory route (Route 1), CE appears at the level of the charges: the per-level marginals are σ-additive, which by Theorem 3.8 is equivalent to CE applied at each level. Here, CE is an *algebraic* condition: the charges do not assign persistent mass to events that the observer's refinement hierarchy sees as empty. It is a condition on the observer's commitments.

In the Stone route (Route 2), CE appears as a *topological* condition: the Baire measure $\hat{\mu}$ on the Stone space must be supported on $\mathrm{pure}(\Omega)$, the image of the sample space inside its compact compactification. Mass on non-principal ultrafilters corresponds exactly to the purely finitely additive part μ_p, and CE $\Leftrightarrow \mu_p = 0$ (Proposition 5.6). Here, CE is a *geometric* condition: the observer's measure does not escape to the "phantom" points of the Stone compactification.

The Yosida–Hewitt decomposition is the bridge between the two faces: it identifies the purely finitely additive part of μ with the mass that has escaped to non-principal ultrafilters. CE says both things simultaneously: the algebraic convergence is honest, and the geometric measure is confined to Ω.

The two routes are therefore not symmetric alternatives. In the Carathéodory route, σ-additivity is assumed from the outset and CE is the condition it amounts to in the present framework. In the Stone route, compactness manufactures σ-additivity on the compact completion, and CE is then the support condition that descends it back to Ω. In both cases CE is doing the same work: ensuring that no mass escapes the realised observational world.

6.3 The Stone space as compact completion

The Stone space $\mathrm{St}(\mathcal{C})$ constructed in §5 as a technical device for measure extension is the canonical compact Hausdorff completion of the sample space for the observable topology. The

Stone embedding pure : $\Omega \to \mathrm{St}(\mathcal{C})$ is dense — every nonempty basic clopen $[E]$ meets pure(Ω) — and the Stone space is uniquely determined by $\mathcal{C}$ up to homeomorphism. It is not imposed on Ω from outside but the compactification already contained in the Boolean algebra of cylinder sets.

Every point in $\mathrm{St}(\mathcal{C})$ is an ultrafilter: a maximal consistent assignment of yes/no answers to every observable distinction. Principal ultrafilters correspond to actual sample-space points; non-principal ultrafilters are the ideal limit points — consistent distinction patterns the refinement system is coherent with, but that no element of Ω realises. The Stone space is therefore the *space of all consistent distinction patterns*: the completion of the observer's world to a compact whole.

CE is the condition that closes the loop. It requires the measure P on Ω, when extended to $\mathrm{St}(\mathcal{C})$ via pure, to remain concentrated on the realised part: no mass escapes to the phantom ultrafilters. An observer satisfying CE retains no phantom mass: probability remains confined to the realised world, however far the algebraic completion extends. The Stone topology, generated by all cylinder clopens, is the coarsest topology making every observable distinction continuous.

For a dynamical system (X, T, μ) with observation function h, the time-delay cylinder algebras form a directed system of exactly the kind treated here. Whether the Stone space of the resulting observable algebra recovers the state space X depends on conditions developed in the companion papers.

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